
***Bahir Dar University IOT School of Civil and Water Resource Engineering
Holistic Exam for Civil Engineering Extension Program***

Date: 15/07/05 E.C

General Instructions:

- *There are totally 50 questions*
- *Depending on the question type choose the **best answer** or give **neat and short** answer for each of the questions.*
- *The exam is closed book.*
- *The allotted exam time is **3& 1/2 hours only**.*
- *Being found in any fraud activity during the exam will result in getting null mark.*

- 1) Discuss the basic classification of surveying

Ans. Plain and geodetic

Plane – flat, geodetic- ellipsoid

- 2) Explain briefly with sketch and formula type of leveling used to determine the height of high rising Structures.

Ans. Trigonometric, reciprocal, inverted

- 3) Explain briefly the use of topographic maps for Civil Engineers.

Selecting route, crossing for bridges, canals and dams

- 4) Discuss the different types of curves. Also discuss their importance.

Horizontal- change the alignment of direction

Vertical- change slope of alignment

Transition- continuously varying radius

Circular- constant r

Compound- two or more radius intersect in the same direction

- 5) List and explain the basic reference systems in surveying both for vertical and horizontal distance measurements.

Geoid, ellipsoid, datum

Datum- a reference which measurements are taken from

Ellipsoid- a mathematical Surface obtained by revolving an ellipse about an earth axis.

Geoid- a reference point when measuring height

- 6) What are the four basic distinct processes in the perception reaction time? And if you are asked to select the driver in a given area; what are your criteria for the selection?

Perception, identification, decision, reaction

- 7) Discuss the challenges for transport planners to make reliable forecast of traffic demand.

Land use, socio economic, availability of transport facility and service

- 8) Among all the modes of transportation road transport is the nearest to the people; write at least five reasons why it is so?

Wide geographical coverage, low cost, door to door service, flexibility

- 9) What are the main design controls and criteria you are going to give attention during the design of a given highway road and write briefly how and when they affect your design.

10) What is CBR? And its significance for civil engineering structures? What is the function of the surcharge disc in CBR test?

CBR test is the most common strength test conducted on soils and other unbounded road making to measure the resistance to penetration of plunger materials. Used to determine the potential strength of sub-grade, sub-base and base course materials, including recycled materials for use in road and airfield pavements.

Surcharge- used to simulate or represent the anticipated pavement layer weight over the material at the site.

11) Chemical weathering occurs because of

- a. temperature changes
- b. spreading of plant roots
- c. crystal growth
- d. carbonation**

12) Expansive soils swell because they contain an abundance of:

- a. kaolinite
- b. illite
- c. montmorillonite**
- d. halloysite

13) Soil can exist in

- a. three phase system
- b. four phase system
- c. two phase system
- d. a and c**

14) obtain grain size distribution, hydrometer analysis is appropriate for

- a. Pebbles
- b. silts and clays**
- c. sands and gravels
- d. all soils

15) Effective stress in a soil is the stress carried by the

- a. Pore water
- b. Pore air
- c. Soil grains**
- d. none of the above

16) A vertical stress of 200 kPa is applied to a saturated clay, the excess porewater pressure, the effective stress and the total stress respectively at $t=\infty$ are in (kPa)

- a. 0,200,200
- b. 200,0,200**
- c. 100,100,100
- d. none of the above

17) The maximum shear stress in case of Mohr's circle will be numerically equal to

- a. 1
- b. 3
- c. $\frac{\sigma_1 + \sigma_3}{2}$
- d. $\frac{\sigma_1 - \sigma_3}{2}$**

18) Which one of the following exhibits more stress during mobilization?

- a. passive resistance
- b. active resistance**
- c. at rest condition
- d. K_0 condition

19) Two footings, one with circular and the other of square are founded on the **surface of cohesion less** soil. The ratio of their ultimate bearing capacities according to Terzaghi is (assume diameter of the circular footing = width of square footing)

- a. 1.0
- b. 2.0
- c. 1.5**
- d. None

20) The extent of ground investigation is determined by:

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- a. The character and variability of the ground.
 - b. Groundwater
 - c. Amount of existing information
 - d. a and c
 - e. all**

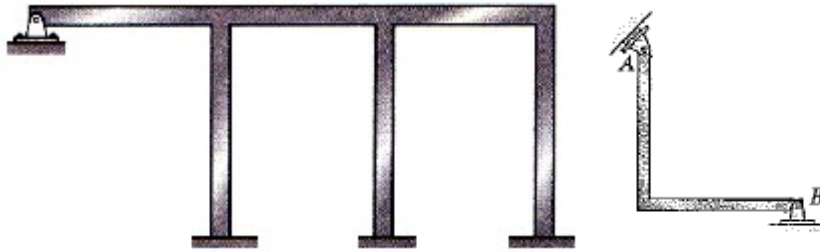
21) If the effective depth of an isolated footing supporting a concrete column is 40cm, the critical sections for moment, wide beam shear and punching shear from the face of the are: (according to EBCS 2-1995)

- a. 80, 40,0 cm respectively
- b. 0,40,80 cm respectively**
- c. 0,80,40 cm respectively
- d. 80,0,40 cm respectively
- e. None

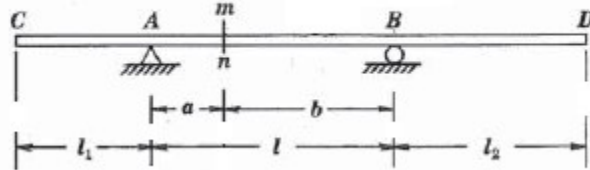
22) Which one of the following is stabilizing force on retaining structures

- a. passive earth pressure
- b. active earth pressure
- c. weight of the retaining structure
- d. all of the above
- e. a and c**

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- 23) Which one of the following is categorized under deep foundation
- a) Pier
 - b) Cassisons
 - c) Piled-raft
 - d) all**
 - e) a and b
- 24) Which type of structures (**light** / heavy) is more susceptible to damage from expansive soils?
- 25) What makes the water to flow through conduit/ pipes? Illustrate with figure.
- 26) What makes weir and barrage different from dam?
- 27) Explain,
- a. The difference between open channel and pipe flows.
 - b. What is hydraulic jump? And explain with sketches what the purpose of studying hydraulic jump is.
- 28) What is buoyancy force and state about the principle behind?
- 29) Explain momentum principle and its application.
- 30) Which of the following type of dam transmits its load by slope stability mechanism?
- a. Gravity dam
 - b. Arch dam
 - c. Buttress dam
 - d. Embankment dam**
- 31) What possible condition of equilibrium of solid body takes place when a small displacement produces an over turning moment tending to displace the body further from its equilibrium position
- a. Stable equilibrium
 - b. Unstable equilibrium**
 - c. Neutral equilibrium
 - d. a and c
- 32) What is the main criterion used to classify open channels as steady or unsteady flow.
- a. space or distance
 - b. time**
 - c. viscosity and friction force
 - d. density
- 33) Determine whether the structures shown on next page are stable or unstable. If they are unstable, draw a picture showing design changes that would make them stable. If they are stable, determine whether they are determinate or indeterminate with respect to restraints. If they are indeterminate, specify the degree of indeterminacy.



34) Draw qualitatively the influence line for shear force and bending moment at section m-n of beam CABD qualitatively.



35) For the proposed beam section, if the check for diagonal compression fails, what will be your recommendation?

- a) Providing appropriate shear reinforcements
- b) Increasing the depth**
- c) Provide longitudinal reinforcements
- d) Increasing the width
- e) Increasing the grade of the steel.

36) All are correct about structural redundancy in reinforced concrete except:

- a) It allows inelastic load redistribution
- b) It ensures structural ductility
- c) It ensures structural strength
- d) It simplifies design process
- e) It ensures economical design

37) For constructions directly exposed to the atmosphere, the most durable structures are:

- a) Steel structures
- b) Masonry structures**
- c) Reinforced structures
- d) Timber structures
- e) All

38) All are true for C-25 concrete works except:

- a) The minimum compressive strength is 25Mpa
- b) The average compressive strength [28 days] of the tested samples should be at least 25Mpa
- c) The maximum compressive strength [28 days] of the tested samples should be at least 30Mpa**
- d) None

39) Which one of the following is false

- a) Due to crack, large portion of reinforced concrete section is not effectively used
- b) In ductile failure of the reinforced concrete sections the steel yields before the concrete crush and there will not be sudden failure.
- c) In reinforced concrete sections where large amount of steel is used, failure will be sudden without warning.
- d) To increase the bond between the concrete and steel using smaller diameter than larger diameter of reinforcement bar with the same area is preferable
- e) None of the above

40) Describe briefly Strength, Stiffness and Stability of a structural member.

41) Draw and describe Stress-strain diagram for a typical structural steel under uni axial tension test.

42) The unfitness of structures for serviceability limit state from deflection can be prevented

- a) By keeping the calculated deflection below the limit values
- b) By compliance with the requirements for minimum effective depth
- c) a& b
- d) none of the above

43) What is the difference between force/flexibility and displacement/stiffness methods of statically indeterminate structures analyses?

44) Which one of the following statement is correct about analysis and design of reinforced concrete columns:

- a) In sway type of frame structure, the column can be designed as isolated column
- b) The minimum number of reinforcement bars is not depending on the cross-sectional shape of the column.
- c) **Even if the frame system is non-sway, the design should include the second order eccentricity.**
- d) The main design action effect is axial force not the moment.

45) What is the function of curing of concrete and list curing methods?

46) How do we test the quality of coarse aggregates? As a Currently admixture is more popular instead of producing different types of Portland cement what is the reason behind?

47) List out and describe at least 4 points that should be fulfilled during design and construction of buildings.

48) What is the difference between structural analysis and design?

49) How can we check consistency of fresh concrete?